Identification of Cryptic Pockets in RAC1 and RHOGAP/GEFs binding interfaces

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Abstract

The work described here attempts to locate cryptic pockets in the RAC1 protein. More to be written

Introduction

- importance of GTPases, their switch mechanism and their interactions with GEFs, GAPs and GDIs
  Small GTPAses act as molecularswitches cycling between OFF and ON states. This switching mechnism thus controls a vast variety of processes that control cell growth and differentiation to vesicular and nuclear transport.
- structural biology information about RAC1 and protein-protein interactions with GEFs, GAPs and GDIs
- Therapeutic interventions on RAC1 and their disadvantages
- Binding site information and changes in the binding sites, if any are reported

- Are there any reports on cryptic pockets for RAC1

Methods

Table 1: Detailed system studies

Initial PDB	System	Size (x, y, z, nb atoms)	Simulation Time (ns)	Replicates
1PDB	Rac1	33x34x22, 29000	250	1
2PDB	Rac1-PREX1	33x34x22, 29000	200	1
3PDB	Rac1-TIAM1	33x34x22, 29000	128	1
4PDB	Rac1-ligand	33x34x22, 29000	129	1
and	others	33x34x22, 29000	tbc	tbc

- List of PDBS used. All systems modelled are described in Table 1.

- Protocol for AMBER MD simulations

  Preparation of systems for molecular dynamics simulation:

  The coordinates for the protein-ligand complexes were retrieved from the Protein Data Bank (PDB); the PDB entries used in this study are listed in Table S1 in the supplementary information. The X-ray structures were imported and processed in the Maestro-GUI (Schrödinger suite, 2018). All systems were prepared for MD simulations with the Leap module in AMBERtools15. Protein atoms were atom-typed according to the AMBER14 (ff14SB) force field.

  Generating AMBER-compatible parameters for ligands:

  Molecular dynamics simulations:

  All complexes were solvated with TIP3P waters22 enclosed in a box with dimensions extending 12 Å beyond any protein atom, and an appropriate number of neutralizing ions were added to maintain the neutrality of the system. The complexes were relaxed using three cycles of minimization of 50,000 steps each. The first 10,000 steps of minimization adopted the steepest descent algorithm, and the remaining 40,000 steps used the conjugate gradient algorithm. The cut-off for evaluating non-bonded interactions was placed at 8.0 Å for the minimization phase. The first cycle of minimization was performed with 25 $kcal/(mol\mathring{A}^2)$ restraining force on the solute atoms, this force constant was reduced to 5 $kcal/(mol\mathring{A}^2)$ in the next cycle, and the final cycle was performed without any restraining forces on either the solute or the solvent atoms. The system was heated from 0 K to 300 K for 1 ns with 5 $kcal/(mol\mathring{A}^2)$ restraining force on the solute atoms using the Langevin dynamics, with a collisional frequency of 2 ps^-1. After heating, the system was equilibrated by MD simulations under the NVT ensemble for six cycles of 1 ns each, where the restraining force on the solute atoms was gradually reduced from 5.0 to 0.0 $kcal/(mol\mathring{A}^2)$. The next step employed the NPT ensemble for density equilibration for another six cycles of 1 ns each, with the restraining force on the solute atoms gradually decreasing from 5 to 0 $kcal/(mol\mathring{A}^2)$. In the equilibration phase, the cut-off for evaluating non-bonded interactions was placed at 9 Å. In the production phase of the MD simulations, all systems were simulated for 100 ns in triplicate, amounting to a 300 ns trajectory for each system. The long-range electrostatic interactions were calculated using the Particle Mesh Ewald (PME) summation method. The minimization and equilibration phases of MD simulations were carried out with the MPI version of the SANDER and PMEMD codes implemented in AMBER22, respectively. The production phase was carried out using the CUDA-enabled code of PMEMD in AMBER22. All simulations were performed with the SHAKE25 algorithm to constrain bonds to hydrogen atoms, with the tolerance set to 0.00001 Å. A 2 fs integration time step was used to solve Newton's equations of motion. For analysis, the conformations were saved every 100 ps, amounting to 1000 snapshots per trajectory (3000 per system).

Analysis of MD trajectories:

Analysis of the MD trajectory was performed with the Cpptraj module in AMBER-

Tools23. Before the analysis, all solvent atoms and neutralizing ions were stripped off, and the protein conformations were aligned to the first frame. The mass-weighted Cα root-mean-square deviation (RMSD) for the protein was computed using the X-ray conformation as the reference. The convergence of MD trajectories was estimated using the root-mean-squared average correlation (RAC) with mass-weighted RMS averaging of Cα atoms, with the first frame of the trajectory as the reference. The root-mean-square fluctuations (RMSF) per residue for the backbone and side-chain atoms were computed from the last 50 ns part of the trajectories.

- Protocol for POVME2
  - pocket identification
  - pocket volume measurements
- fragment placement and trying yo keep the pocket open to target drug-like molecules
  - Type of pocket and the type of fragment more suitable to keep it open
- Analysis of other MD observations to support pocket volume changes
  - RMSDs for switch region 1 and 2
  - (perhaps) PCA analysis to explain the switch region dynamics and the pocket volumes

Results and discussion

We studied how different Rac1 experimental structures were comparable to each other by computing their RMSD. This allowed to select X structes in the apo form of Rac1, then Y structures with Rac1 bound to one partner, Z structures where Rac1 was bound to a ligand, and S structures where RAc1 was both bound to a partner protein and a ligand.

- Try to look for a connection/relation between the following
  - switch region dynamics with different binding partners
  - switch region dynamics with pocket volumes
  - If we find different pockets that might open and close with different binders

Discussion

Proteins are dynamics by nature, even if RAs members are closely related, their subtle sequence differences are linked to specific

Figures/image1.png

Figure 1: Systems prepared and studied in this article

Figures/pocketEvolution.png

Figure 2: Pocket evolution over time

Figures/image1.png

Figure 3: Conservation of pocket in simulations. (A) Pockets found in all prepared system. (B) Pockets specific for PREX1. (C) Pockets specific for TIAM1.

Figures/image1.png

Figure 4: Binding energy of selected ligands in their pockets, and detailed energetics contributions using MMGBSA.

Figures/pocketAnalysis.png

Figure 5: Amino acids involved in pocket definition, and importance of their role in ligand binding. AKA ligand footprint in the pocket of interest.